

Database Assignment 10

Q1) Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).

```
w1_99099_vaibhavi>DELIMITER $$
w1_99099_vaibhavi>
w1_99099_vaibhavi>CREATE PROCEDURE ComputeRectangleProperties()
-> BEGIN
->
->     DECLARE L INT DEFAULT 10;
->     DECLARE W INT DEFAULT 5;
->
->     DECLARE Perimeter INT;
->     DECLARE Area INT;
->
->     SET Perimeter = 2 * (L + W);
->     SET Area = L * W;
->
->     SELECT L AS Length, W AS Width, Perimeter AS Perimeter, Are
a AS Area;
-> END;
-> $$
Query OK, 0 rows affected (0.01 sec)

w1_99099_vaibhavi>
w1_99099_vaibhavi>DELIMITER ;
w1_99099_vaibhavi>CALL ComputeRectangleProperties();
+-----+-----+-----+-----+
| Length | Width | Perimeter | Area |
+-----+-----+-----+-----+
|      10 |      5 |          30 |      50 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

w1_99099_vaibhavi>
```

Q2) Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
w1_99099_vaibhavi>CREATE PROCEDURE calc_cube(IN num int)
-> BEGIN
->     DECLARE cub int DEFAULT 0;
->     DECLARE sq int DEFAULT 0;
->
->     SET sq = num * num;
->     SET cub = num * sq;
->
->     SELECT num,sq AS square,cub AS calc_cube;
->
->     INSERT INTO TEMP VALUES(num,sq,cub);
-> END
-> $$
Query OK, 0 rows affected (0.01 sec)

w1_99099_vaibhavi>DELIMITER ;
w1_99099_vaibhavi>CREATE TABLE TEMP (VALUE int, SQUARE int, CUB int );
Query OK, 0 rows affected (0.03 sec)

w1_99099_vaibhavi>SELECT * from temp;
Empty set (0.00 sec)

w1_99099_vaibhavi>INSERT INTO temp (VALUE, SQUARE, CUB) VALUES (5, 25, 125);
Query OK, 1 row affected (0.01 sec)

w1_99099_vaibhavi>select * from temp;
+-----+-----+-----+
| VALUE | SQUARE | CUB |
+-----+-----+-----+
|      5 |      25 |  125 |
+-----+-----+-----+
1 row in set (0.00 sec)

w1_99099_vaibhavi>
```

Q3) Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
w1_99099_vaibhavi>DROP PROCEDURE IF EXISTS temp;
Query OK, 0 rows affected, 1 warning (0.00 sec)

w1_99099_vaibhavi>
w1_99099_vaibhavi>DELIMITER $$
w1_99099_vaibhavi>CREATE PROCEDURE temp(IN num DOUBLE)
-> BEGIN
->
->     DECLARE fren DOUBLE DEFAULT 0.0;
->     DECLARE cel DOUBLE DEFAULT 0.0;
->
->     SET fren = (num * 9/5) + 32;
->     SET cel = (num - 32) * 5/9;
->
->     SELECT num,fren AS fahrenheit, cel AS celsius;
-> END;
-> $$
Query OK, 0 rows affected (0.01 sec)

w1_99099_vaibhavi>DELIMITER ;
w1_99099_vaibhavi>CALL temp(40);
+-----+-----+-----+
| num  | fahrenheit | celsius |
+-----+-----+-----+
| 40   | 104       | 4.444444444444445 |
+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

w1_99099_vaibhavi>
```

Q4) Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
w1_99099_vaibhavi>DROP PROCEDURE IF EXISTS inch;
Query OK, 0 rows affected, 1 warning (0.00 sec)

w1_99099_vaibhavi>
w1_99099_vaibhavi>DELIMITER $$
w1_99099_vaibhavi>CREATE PROCEDURE inch (IN num DOUBLE)
-> BEGIN
->     DECLARE inch DOUBLE DEFAULT 0.0;
->     DECLARE feet DOUBLE DEFAULT 0.0;
->     DECLARE yard DOUBLE DEFAULT 0.0;
->
->     SET inch = num * 12;
->     SET feet = num * 3;
->     SET yard = num / 36;
->     SELECT num,inch AS inch,feet AS feet,yard AS yard;
->
-> END;
-> $$
Query OK, 0 rows affected (0.01 sec)

w1_99099_vaibhavi>DELIMITER ;
w1_99099_vaibhavi>CALL inch(70);
+-----+-----+-----+-----+
| num  | inch | feet | yard |
+-----+-----+-----+-----+
| 70   | 840  | 210  | 1.9444444444444444 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

Q5) Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
w1_99099_vaibhavi>DROP PROCEDURE IF EXISTS greater;
Query OK, 0 rows affected, 1 warning (0.01 sec)

w1_99099_vaibhavi>
w1_99099_vaibhavi>DELIMITER $$
w1_99099_vaibhavi>CREATE PROCEDURE greater(IN num1 int, IN num2 int)
-> BEGIN
->     DECLARE result int DEFAULT 0;
->
->     SET result = num1 * num2;
->
->     if (result >= 100) THEN
->         SELECT num1,num2,result AS product, 'The product is greater t
han 100' AS message;
->     ELSE
->         SELECT num1,num2,result AS product, 'The product is less than
or equal to 100' AS message;
->     END IF;
->
-> END;
-> $$
Query OK, 0 rows affected (0.01 sec)

w1_99099_vaibhavi>DELIMITER ;
w1_99099_vaibhavi>CALL greater(10,15);
+-----+-----+-----+-----+
| num1 | num2 | product | message                                     |
+-----+-----+-----+-----+
|   10 |   15 |    150 | The product is greater than 100          |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

w1_99099_vaibhavi>
```